

Xin Yu

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EDUCATION

The Pennsylvania State University

PhD Candidate in Statistics, GPA: 4.0/4.0

Aug. 2023 – May 2027 (Expected)

State College, PA

Nankai University

Bachelor of Science in Statistics, GPA: 3.90/4.0

Aug. 2019 – May 2023

Tianjin, China

TECHNICAL SKILLS

- **Foundation Model Training:** RL post-training (GRPO), SFT, CPT, OPD, OPSD, PEFT
- **Agentic AI:** Tool-augmented reasoning, self-evolving agents, critic reward model training
- **Systems:** PyTorch, VERL, FSDP, distributed training, large-scale model training
- **Programming:** Python (PyTorch), R, MATLAB, C++, Java, SQL

EXPERIENCE

Google

Student Researcher

Aug. 18, 2026 – Nov. 17, 2026

Sunnyvale, CA

- Improved Google's internal **cowork coding agent** through **agent post-training** methods based on **Self-Reinforce** and **Recursive Self-Improvement**, strengthening its self-improvement capabilities across the **data agent**, **evaluation harness**, and **training pipeline**.

Meta, Feed Discovery

Research Scientist Intern

May 2026 – Aug. 2026

Menlo Park, CA

- Built a **tool-use-augmented reasoning framework** for ML development by curating a tool library and enhancing small models' **agentic reasoning** and execution capabilities through **agentic RL**.
- Deployed a new generative recommendation generator for the Facebook CCS product line, training it with **CPT** and **SFT** and further improving it with **GRPO** using latent-space reward signals derived from user interactions.

TikTok Ads Integrity

Research Scientist Intern

Nov. 2025 – May 2026

San Jose, CA

- Built an **MLLM-based pipeline** to extract structured SKU attributes from product landing-page screenshots, enabling scalable product understanding for ads integrity and downstream recommendation.
- Improved hard cases through iterative training on failure examples, including **self-rewarding** and **on-policy RL distillation**, and shipped updated checkpoints to production.

Bytedance Seed

Research Scientist Intern

May 2025 – Dec. 2025

Seattle, WA

- Optimized the volcengine/verl reinforcement learning framework with **GRPO in FSDP**, enabling lightweight and high-performance training across **large-scale models**.
- Developed annotation methods for ranking structure in open-source pairwise preference datasets and optimized diffusion models with **list-wise human preferences**.

Microsoft Research Asia

Intelligent Multimedia Group Intern

Jan. 2023 – May 2023

Beijing, China

- Developed lightweight convolutional architectures for visual recognition under data imbalance, improving model efficiency and generalization.
- Shipped the model to an internal OCR pipeline, enabling practical deployment for real-world OCR tasks.

PUBLICATIONS

ExpertWeaver: Unlocking the Inherent MoE in Dense LLMs with GLU Activation Patterns

Ziyu Zhao, Tong Zhu, **Xin Yu**, Zhi Zhang, Tiantian Fan, Jinluan Yang, Kun Kuang, Zhongyu Wei, Fei Wu, Yu Cheng
Forty-third International Conference on Machine Learning (ICML 2026)

Towards Better Optimization for Listwise Preference in Diffusion Models

Jiamu Bai*, **Xin Yu***, Meilong Xu, Weitao Lu, Xin Pan, Kiwan Maeng, Daniel Kifer, Jian Wang, Yu Wang
International Conference on Learning Representations (ICLR 2026)

Each Rank Could Be an Expert: Single-Ranked Mixture of Experts LoRA for Multi-task Learning

Ziyu Zhao, Yixiao Zhou, **Xin Yu**, Zhi Zhang, Didi Zhu, Tao Shen, Zexi Li, Jinluan Yang, Xuwu Wang, Jing Su, Kun Kuang, Zhongyu Wei, Fei Wu, Yu Cheng
SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026)

AltLoRA: Towards Better Gradient Approximation in Low-Rank Adaptation with Alternating Projections

Xin Yu, Yujia Wang, Jinghui Chen, Lingzhou Xue
Neural Information Processing Systems (NeurIPS 2025)

Understanding the Statistical Accuracy-Communication Trade-off in Personalized Federated Learning

Xin Yu, Zelin He, Ying Sun, Lingzhou Xue, Runze Li
International Conference on Machine Learning (ICML 2025)

PREPRINT

MLToolBench: Learning Tool-Augmented Agents for Machine Learning Development

Xin Yu*, Lizhu Zhang*, Jiamu Bai*, Yanhong Wu, Zellux Wang, Serena Li, Weiwei Li, Lingzhou Xue, Xiangjun Fan, Bo Peng
arXiv preprint arXiv:2609.36679, 2026. * Equal contribution.

A New Inexact Manifold Proximal Linear Algorithm with Adaptive Stopping Criteria

Zhong Zheng, **Xin Yu**, Shiqian Ma, Lingzhou Xue
Minor revision informs optimization

PrunedLoRA: Efficient and Robust Gradient-Based Structural Pruning for Low-Rank Adaptation

Xin Yu, Cong Xie, Ziyu Zhao, Tiantian Fan, Lingzhou Xue, Zhi Zhang
Submitted to NeurIPS 2026

EXACT: Explicit Attribute-Guided Decoding-Time Personalization

Xin Yu, Hanwen Xing, Lingzhou Xue
Submitted to NeurIPS 2026

Preference-Based Self-Distillation: Beyond KL Matching via Reward Regularization

Xin Yu, Liucheng Liao, Yiwen Zhang, Yingchen Yu, Lingzhou Xue, Qinzhen Guo
Submitted to NeurIPS 2026

AWARDS

- Machine Learning Summer School Travel Award, Columbia University, Jun. 19–27, 2026
- Best Paper, C.R. Conference, 2025
- Jack and Eleanor Pettit Scholarship in Science, 2024–2025
- University Graduate Fellowship, 2023–2024
- Second Prize, National Undergraduate Academic Competition, 2023
- Second Prize, National High School Mathematics Competition, 2018
- Second Prize, National High School Physics Competition, 2018

PROFESSIONAL SERVICE

- **Reviewer:** ICML 2026 (6 papers), NeurIPS 2026 (4 papers), NeurIPS 2026 Workshop on Meta-Agents (2 papers), AAAI 2027 (4 papers), ECCV 2026 (2 papers), and KDD AI4Sciences 2026–2027 (5 papers)